MANCHESTER 1824
The University of Manchester

AERO60492 – Autonomous Mobile Robots

Lecture 13 – Outdoor Aerial Sensors

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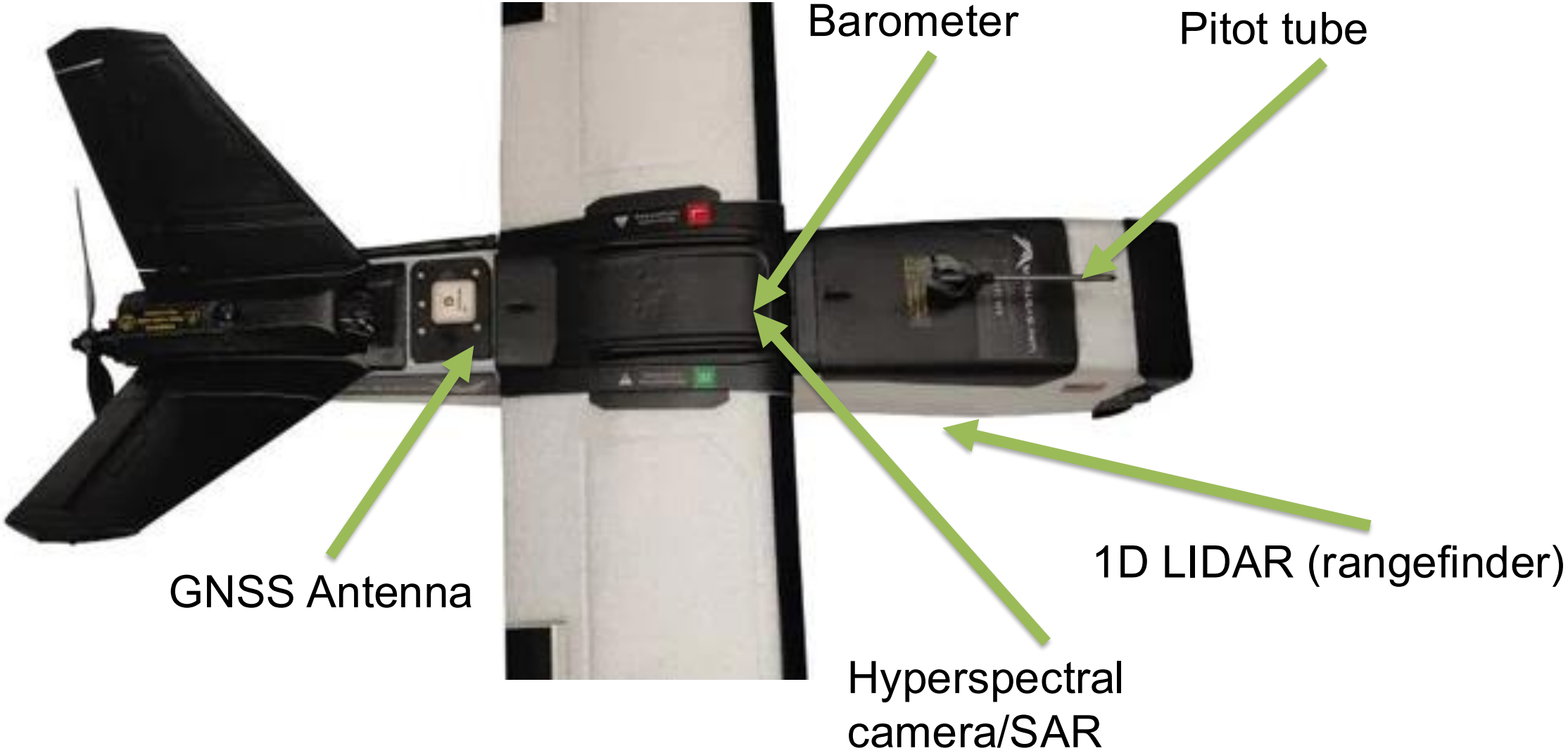
Today’s lecture

- 40 minutes lecture
- 10 minutes quiz
- 10 minutes break
- 40 minutes lecture
- 10 minutes quiz
- End

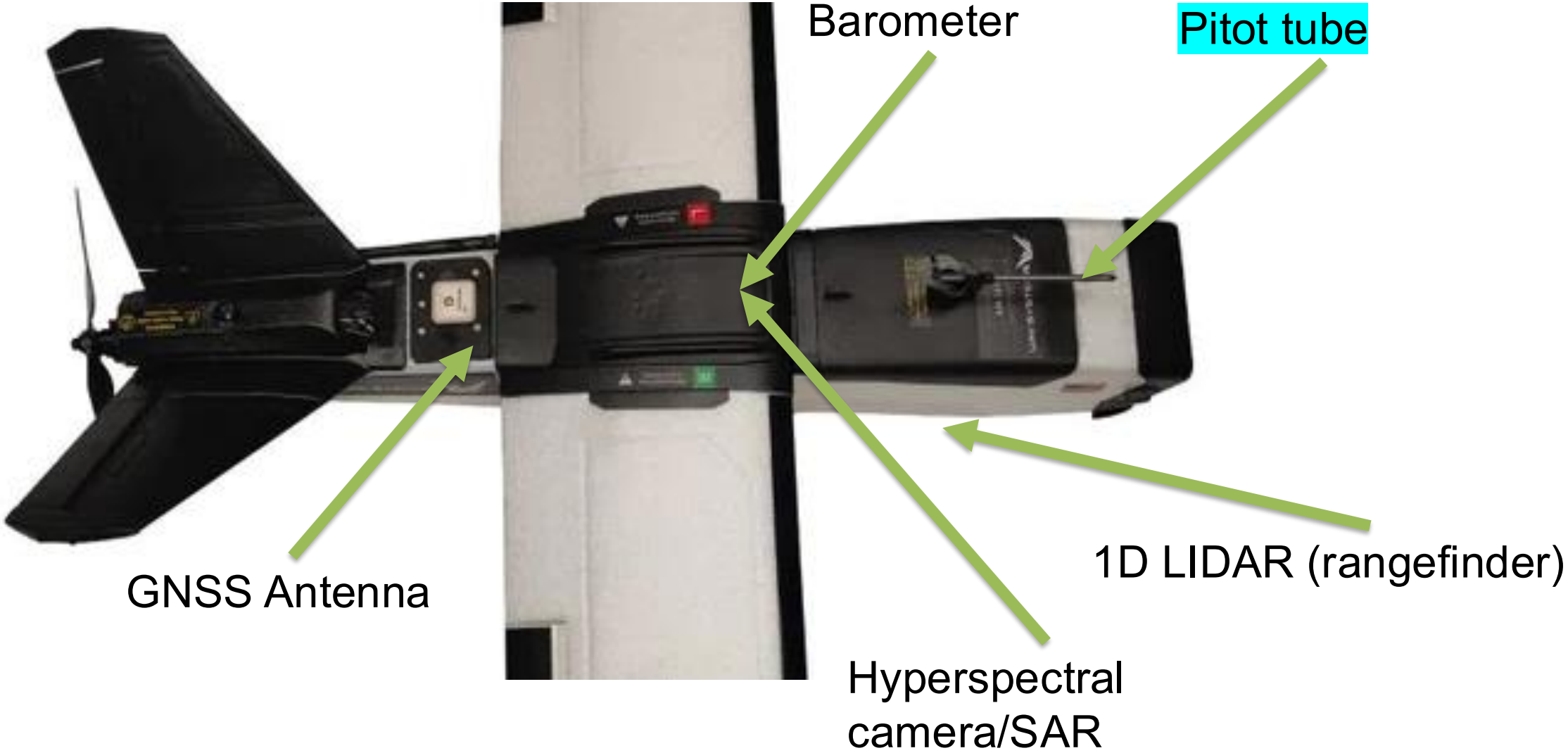
Intended learning outcomes

- Understand what sensors are used on outdoor flying robots
- Explain what pitot tube does
- Understand GNSS working principle
- Explain source of noise and inaccuracies in GNSS
- Appreciate what is Real time kinematic (RTK) GNSS and its limitations
- Appreciate existence of altitude sensors
- Understand the role of SAR and hyperspectral camera

Fixed wing UAV (robot)

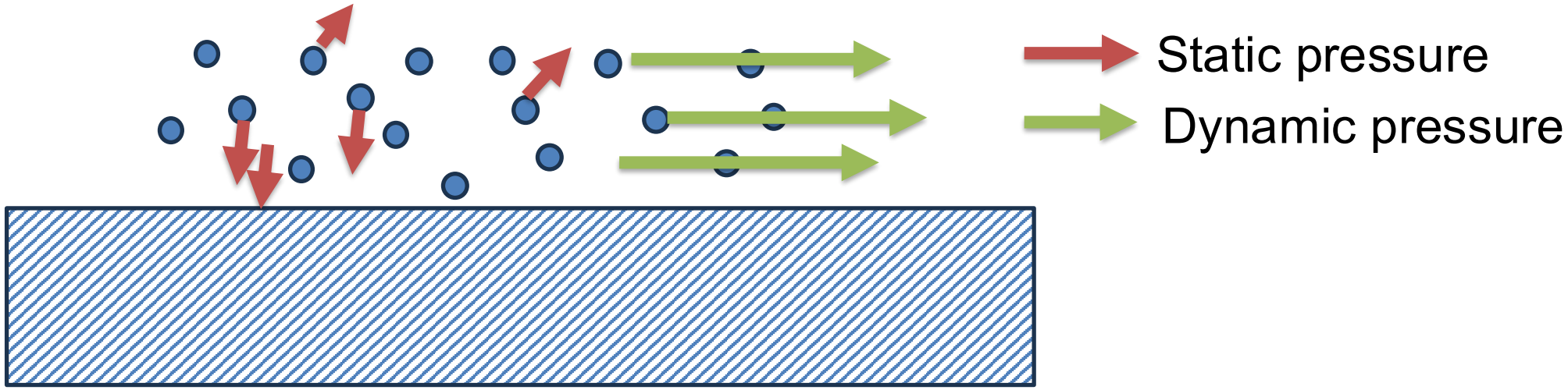


Fixed wing UAV (robot)

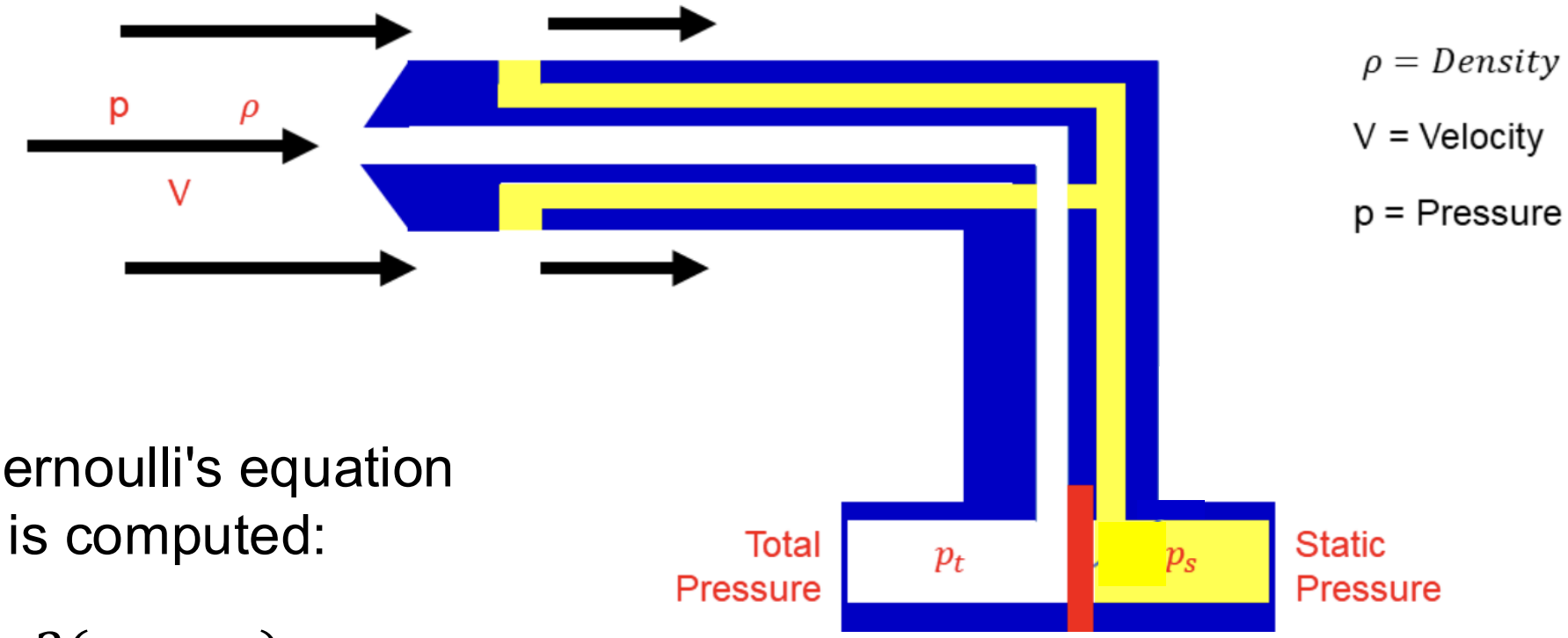


Static vs dynamic pressure (greatly simplified)

- Static pressure is a pressure of a fluid at rest
- Dynamic pressure is pressure of fluid at motion
- Total Pressure is sum of the two
- Intuitive example



Pitot tube explained



$\rho = Density$

V = Velocity

p = Pressure

Using Bernoulli's equation velocity is computed:

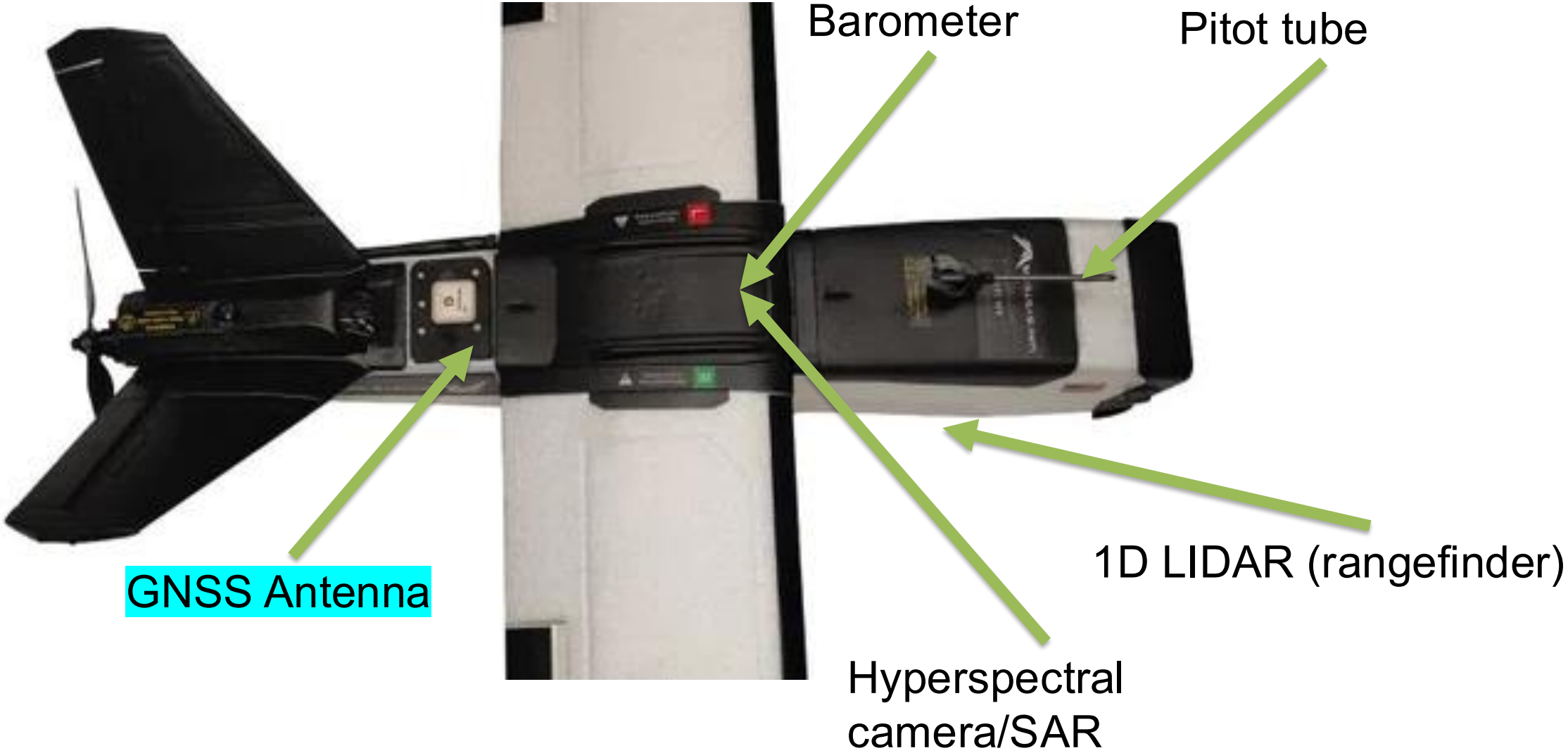
$$V^2 = \frac{2(p_t - p_s)}{\rho}$$

Pressure sensor membrane – computes difference between two pressures

Pitot tube explained 2

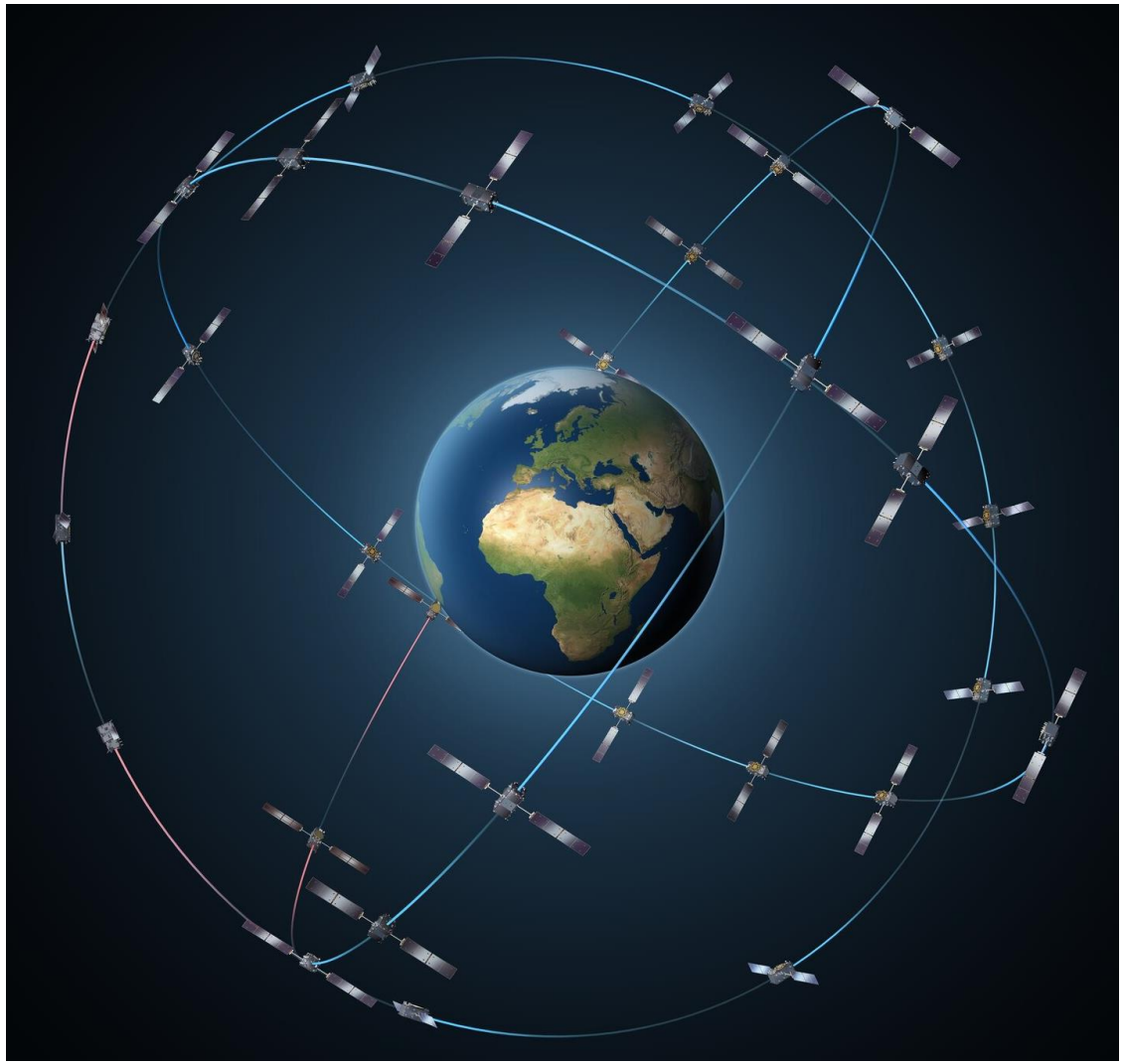
- Pitot tube tells us velocity relative to oncoming air, not the ground!
- In strong wings airplane can still fly but not move relative to the ground (or even move backwards!)

Fixed wing UAV (robot)



Global navigation satellite system (GNSS) introduction

- Allow us to determine position on earth with high accuracy (1-5 meters)
- Oldest system operates since early 90s
- Concept originated as early as 1957 (first satellite)
- Several constellations:
  - GPS
  - Galileo
  - BeiDou
  - …

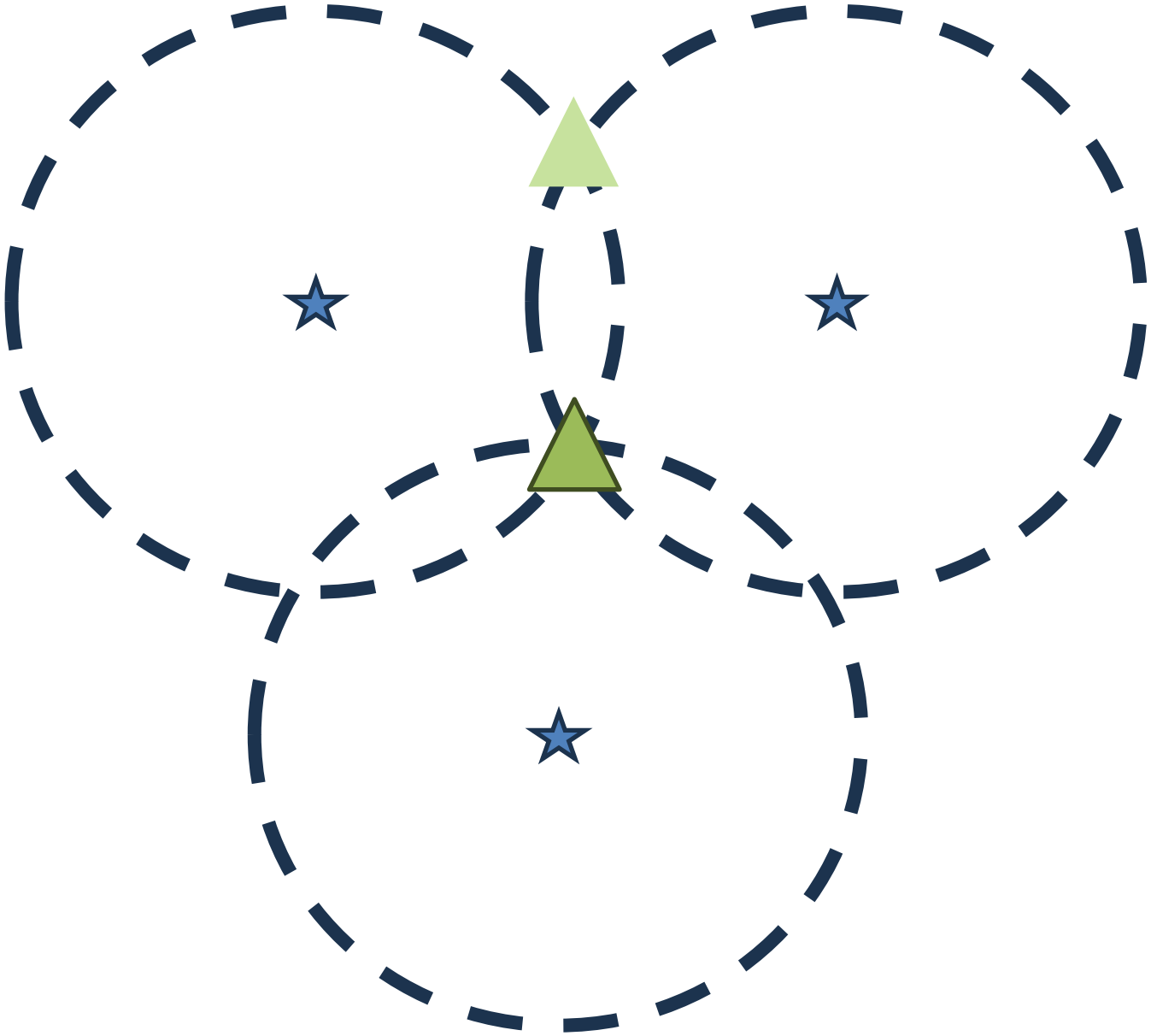


GNSS working principle

- Working principle is not too dissimilar from LIDAR! – still relies on time of flight of signal
- As the return signal cannot be sent (easily), the signal emission time is embedded into the signal
- Positions of satellites are known and also embedded into the sent signal
- Thus, with known flight time and position of the satellite, the distance between receiver and satellite can be computed

GNSS working principle part 2

Satellite
UAV



GNSS source of uncertainties

- The estimated position is affected by following errors
  - Clock errors
  - Orbit errors
  - Ionospheric and tropospheric delays
  - Multipath error
  - Receiver noise
- Thus, more than 4 satellites is required for position estimate, normally at least 7-10 is required

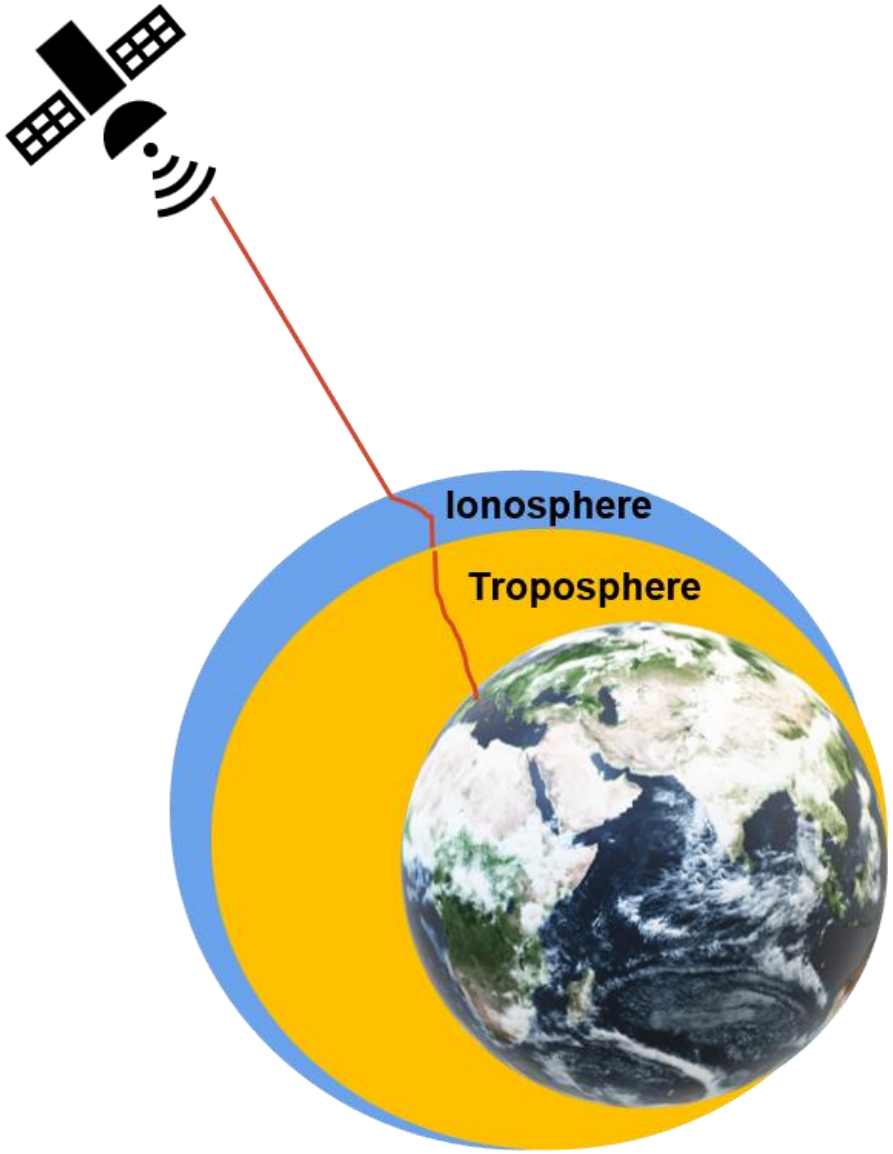
GNSS clock errors

- We measure over significant distance ~20,000 km
- Two clocks are needed: one at the satellite to compute send time, one at the receiver device to compute receive time
- Both need to be extremely accurate,
  - device side can use external source (over the internet)
  - Still some errors are present

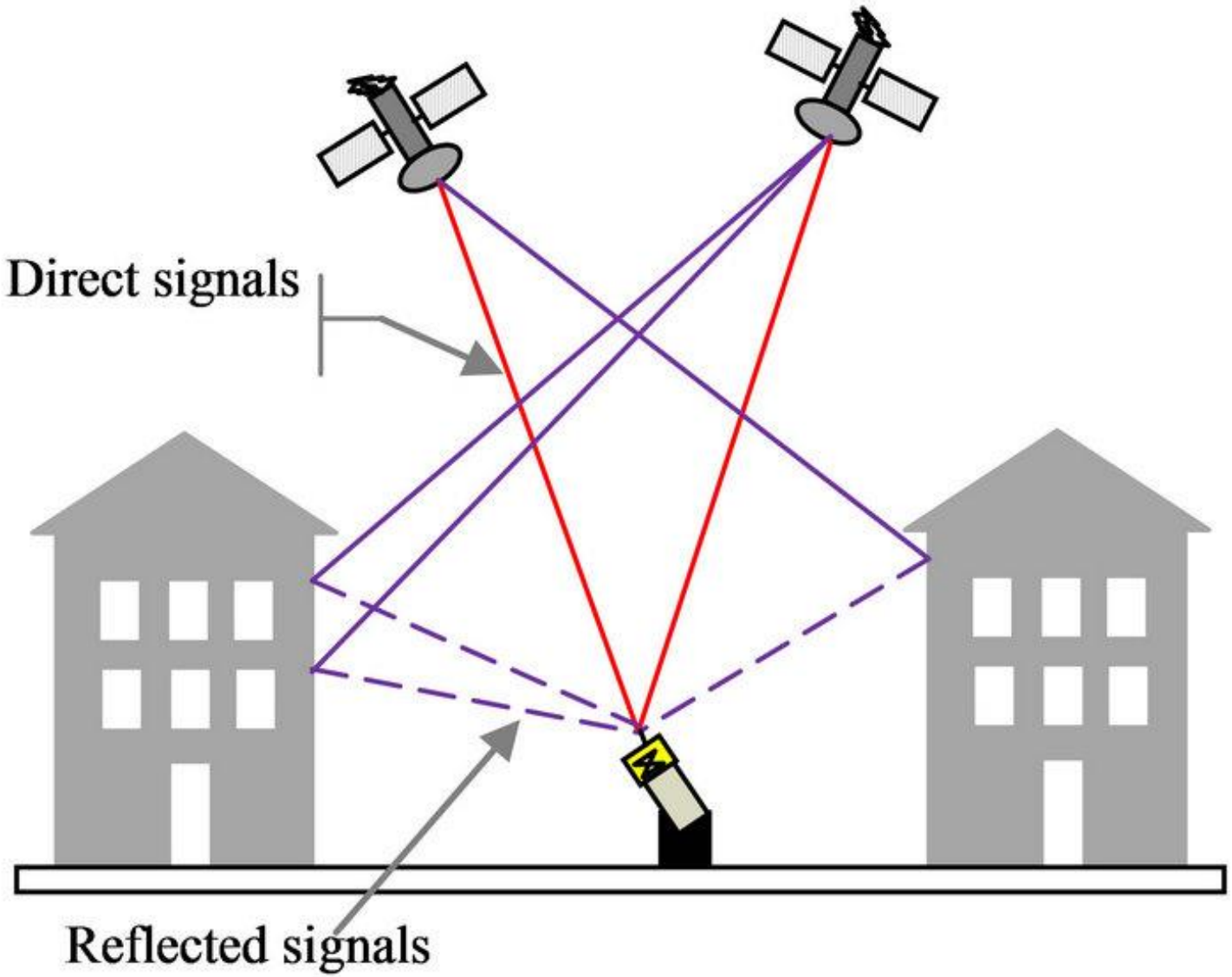
GNSS orbit errors

- Satellite orbit is also not fully known
- It is estimated based on observations and orbital mechanics, but those carry small errors on their own

Ionospheric and tropospheric delays



GNSS multipath error



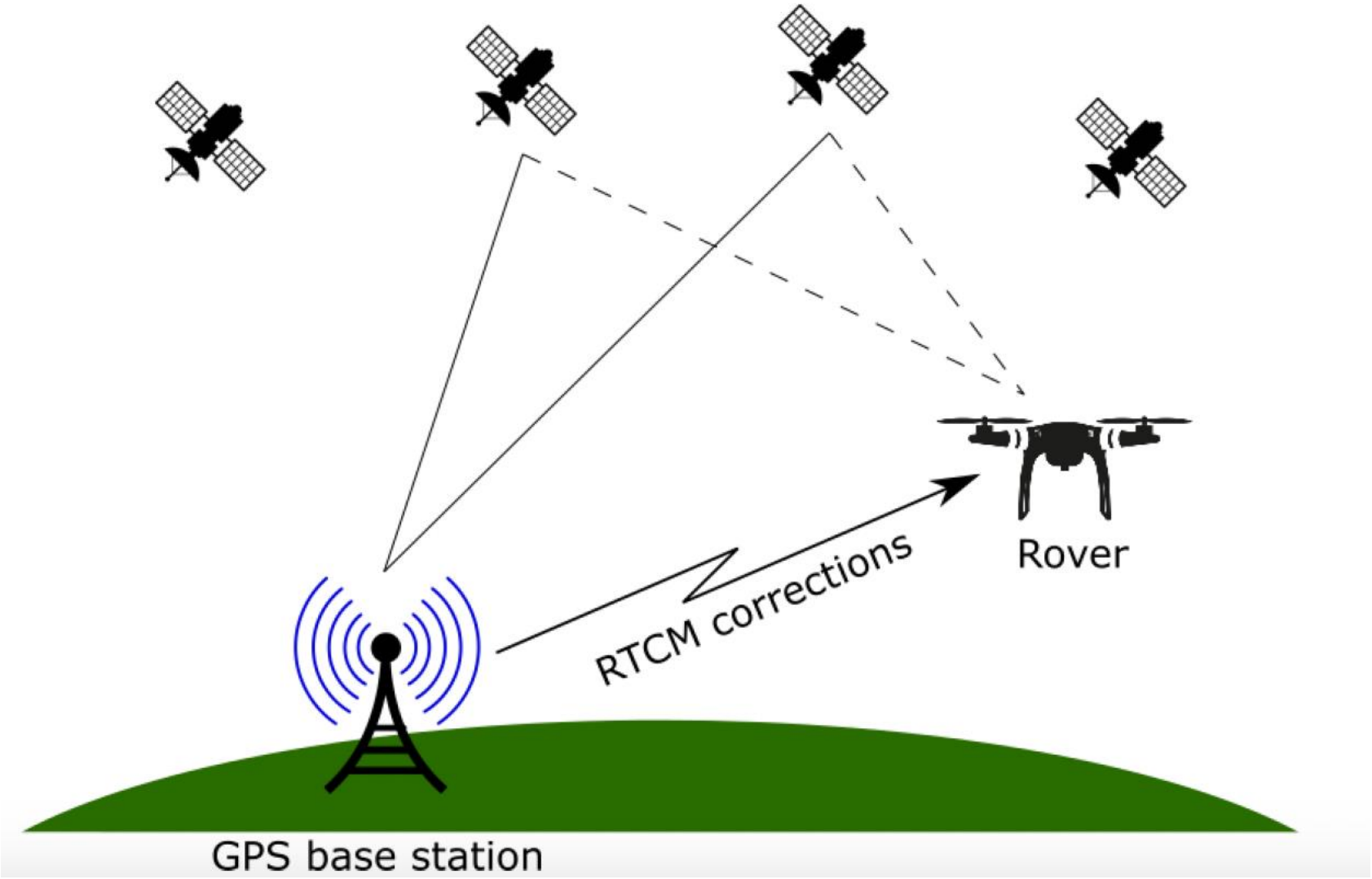
Receiver noise errors

- Wireless signal is susceptible for being affected by other signals
- GNSS signal is not exception
- Often called ‘background noise’ – they come from other processes around the receiver (either accidental or purposeful)

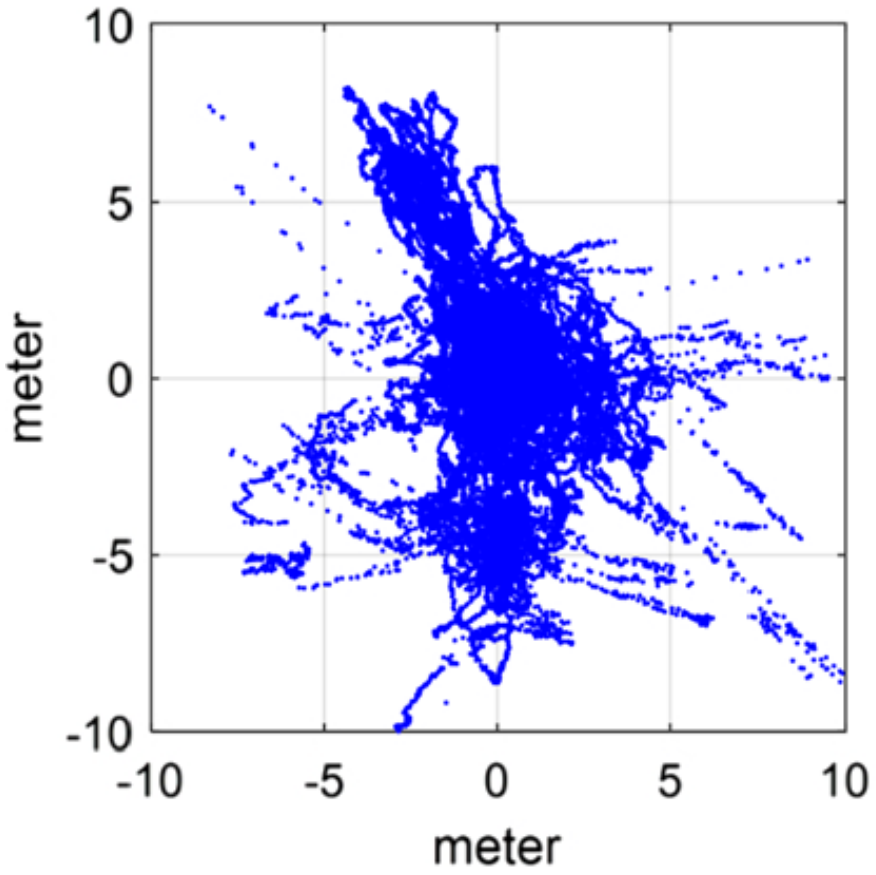
Real time kinematics (RTK) GNSS

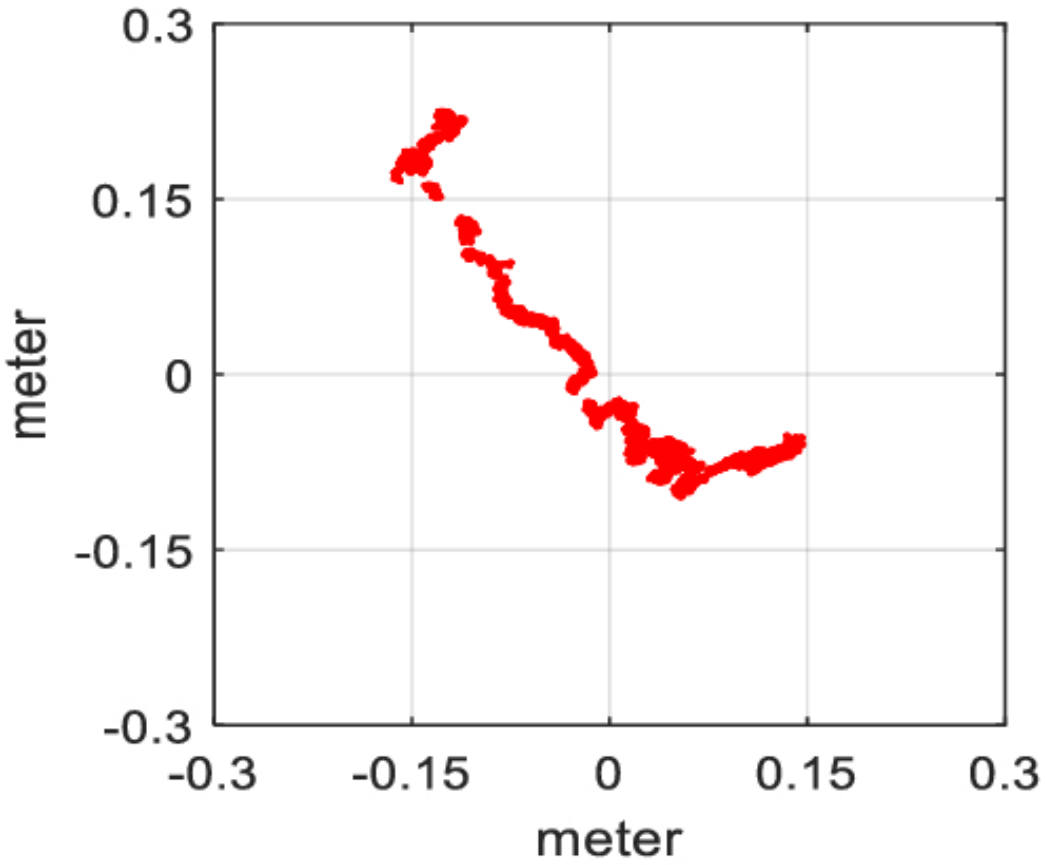
- RTK GNSS utilizes a base station of known position to correct GNSS signal
- Base should be no more than 10 km away
- Base station can be deployed on site
- It gives position relative to base station
- If deployed, base station needs about 10 minutes to fix its own position
- Intuitively (although not strictly correct!): If we know the error in position of the base station, we can extrapolate to the error in nearby robots

RTK GNSS



RTK GNSS results

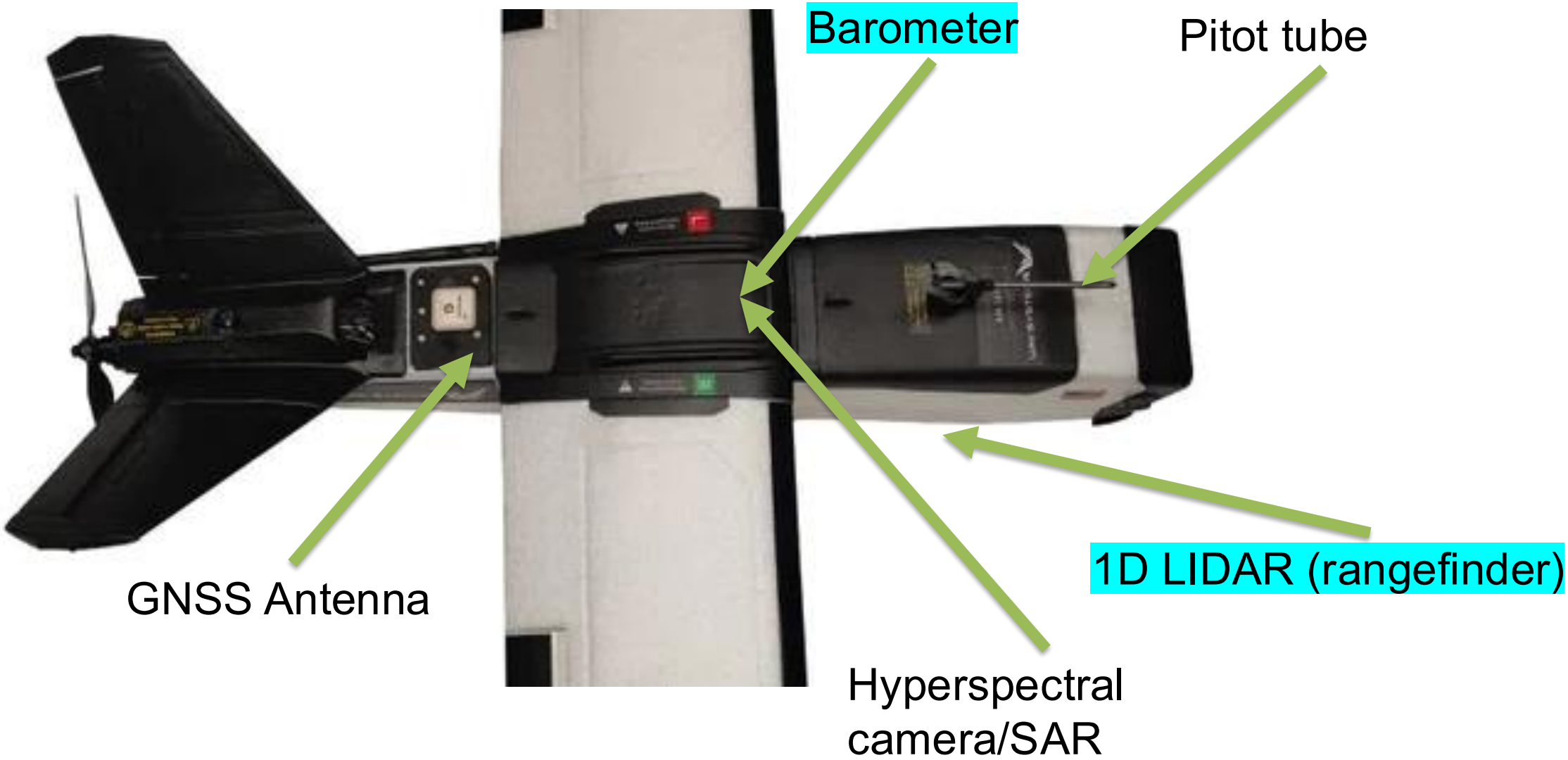




RTK GNSS limitations

- Requires deployment of/subscription to a base station
- Lock can take long time
- If base station position is inaccurate, position of the vehicle (relative to earth reference) is also inaccurate
- Although theoretically 5 satellites is enough, normally requires more than 20 satellites for accuracy

Fixed wing UAV (robot)



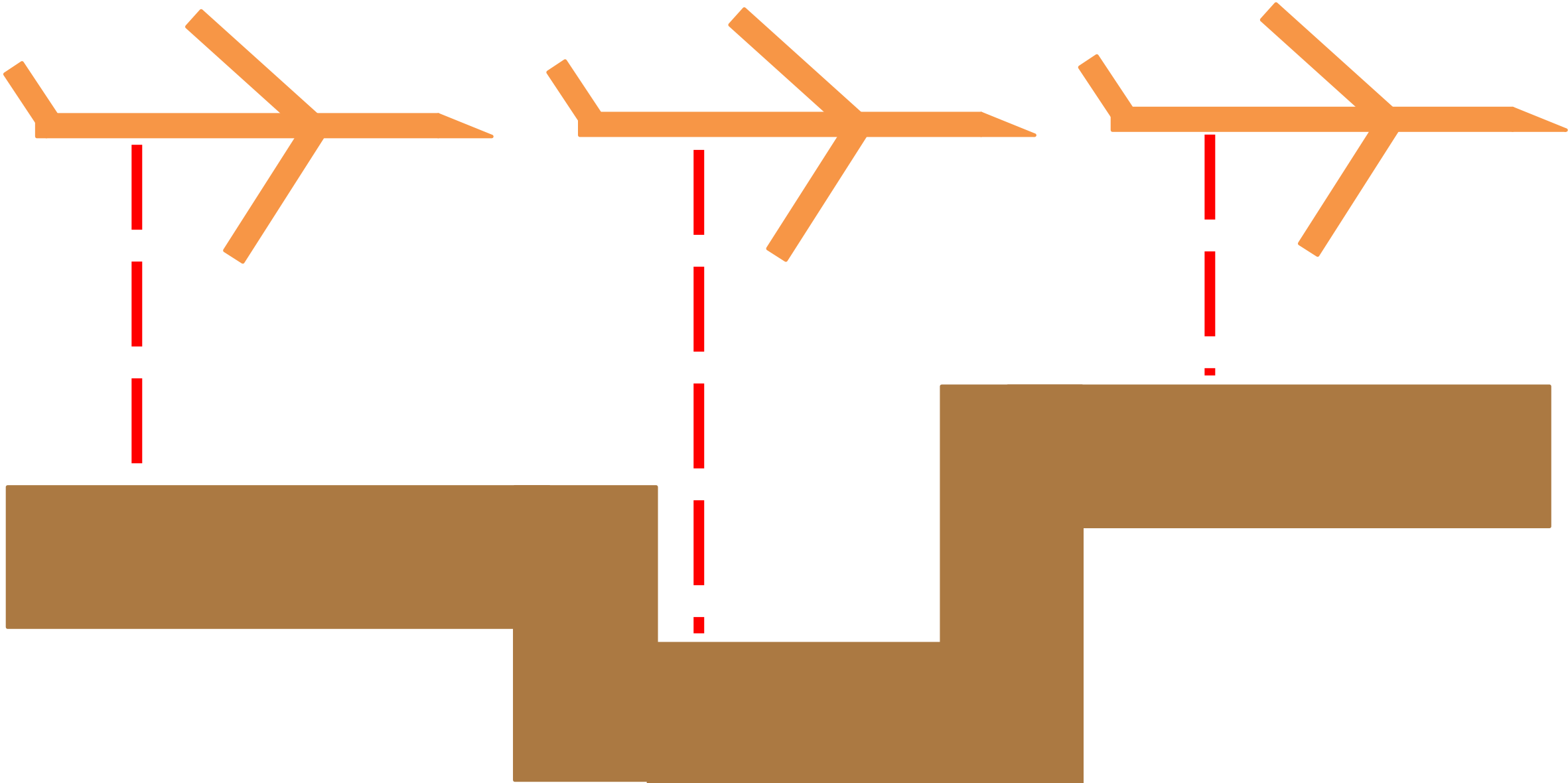
Height Sensors

- GPS height is normally poor accuracy (it is really designed to provide position in x-y plane)
- Other solution
  - Rangefinders
  - Barometers

1D LIDAR (rangefinder)

- Measures distance to the ground below
- Needs to be mounted very precisely
- Very accurate
- issues
  - UAV pitches up or down
  - sudden terrain changes

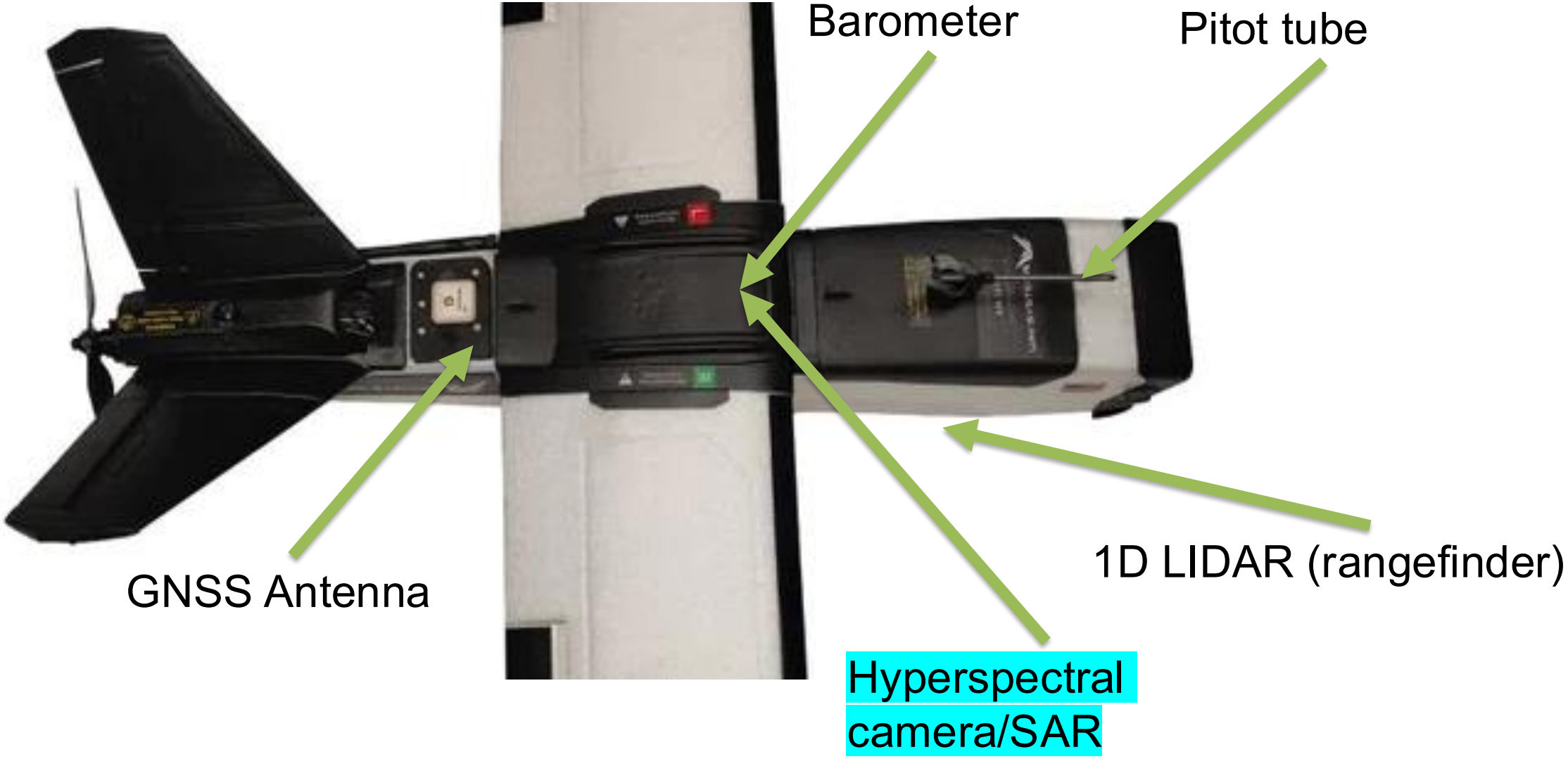
Sudden terrain changes



Barometer

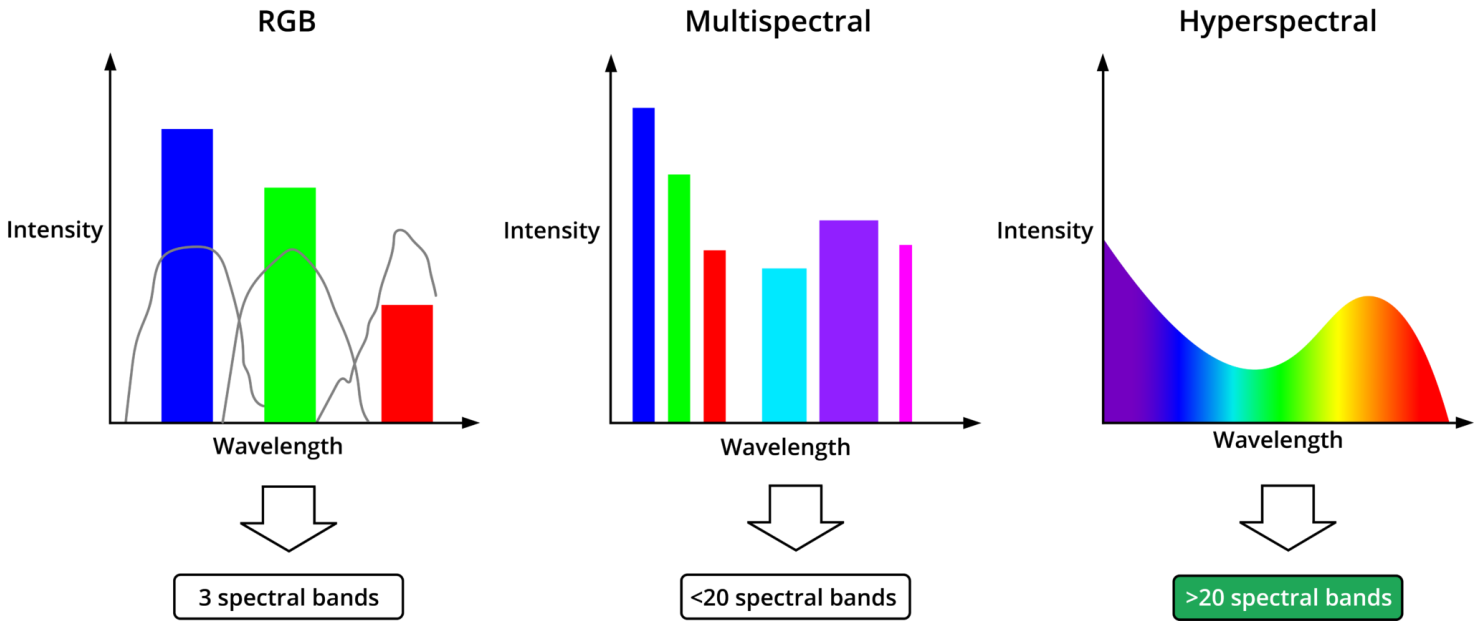
- Converts pressure to height
- Normally part of autopilot hardware
- Does not suffer from aforementioned issues of rangefinders, however not as accurate
- Sensor fusion with rangefinders to get benefits of both

Fixed wing UAV (robot)



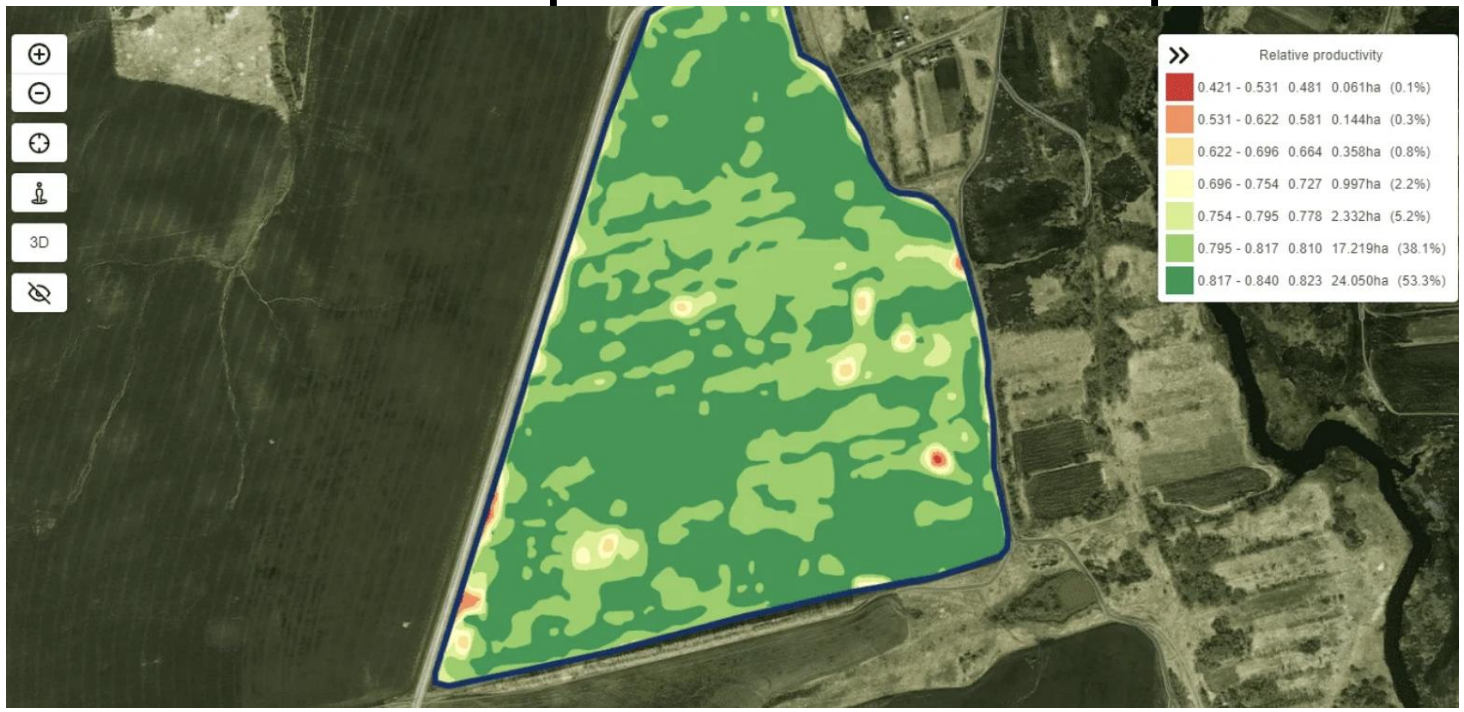
Hyperspectral cameras

- Normal camera works in visible light wavelength
- More information could be available in other bandwidth
- Hyperspectral cameras can observe multiple electromagnetic wavelength bands at the same time
- We normally represent spectrum as various colors



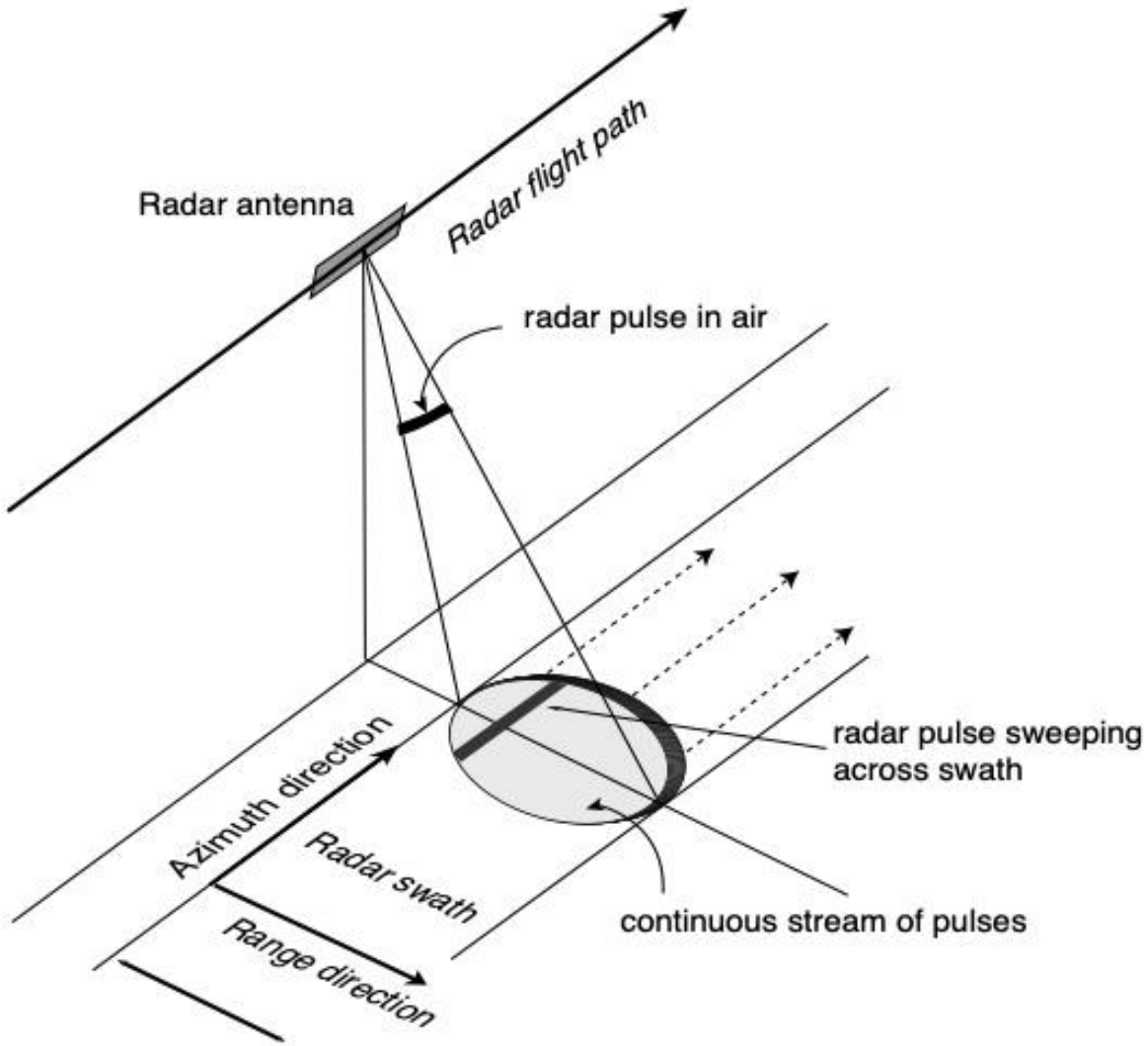
Hyperspectral cameras application

- Agriculture: health, growth, nutrients and water deficiency
- Pharmaceutical: recognize ingredients of pills for production inspection
- Oil and mineral exploration: find deposits



Synthetic Aperture Radar

- Imagine technique which uses pulses of radar to create images
- Works well even in cloudy and/or rainy water
- Can penetrate underground (depends on the wavelength used)
- Generally, not human understandable and heavy post-processing is needed



Applications

- Create maps of caves from air
- Detect underground features such as metal pipes and water streams (note: does not work on PVC pipe unless there is water in them)
- Can see regardless of the weather conditions (very useful on satellites)
- Deforestation monitoring – great for environmental monitoring and control

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Summary

- Sensors used on outdoor aerial robots were introduced
- GNSS system was discussed
- Real time kinematics corrections were presented
- Different height (altitude) sensors were considered
- Hyperspectral cameras were discussed
- SAR was (briefly) discussed

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Quiz